Covariant Biquaternionic Tetrads in UBT: Metric Rank, Connection Reconstruction, and Integrability

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Abstract

Unified Biquaternion Theory (UBT) is a research programme whose canonical starting point is a single biquaternion-valued field $\Theta(q,\tau) \in \mathbb{C}\otimes\mathbb{H}$ over complex time $\tau = t + i\psi$. This paper isolates the local classical gravitational sector of that programme and formulates its geometry directly from the covariant first derivative of Θ. Defining $E_\mu = \mathcal{N}_0^{-1/2} D_\mu\Theta$, the classical Lorentz sector is the real four-dimensional slice $E_\mu = i{e_\mu}^0\mathbf{1} + {e_\mu}^k\mathbf{e}_k$. Quaternion conjugation gives the exact central identity

$$\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1},$$

so the metric is not defined by a trace, real-part map, phase projection, or imaginary-time fiber average. The tetrad-to-metric map has rank ten at every nondegenerate tetrad and a six-dimensional kernel equal to local Lorentz frame freedom. We then reconstruct the connection. For every tetrad and specified torsion, the metric-compatible Lorentz connection is uniquely $\omega = \mathring{\omega}(e) + K(T)$; the torsion-free GR branch is the special case $K = 0$. Hence the remaining full-UBT question is the dynamics of torsion, not a kinematically arbitrary Ω_μ. We give an explicit affine field Θ_{SR} generating the Minkowski tetrad and solving the standard flat source-free second-order master operator. Finally, we prove that a naive one-sided connection with invertible Θ has a flatness obstruction under the stated torsion-free compatibility assumptions, while a two-sided biquaternionic derivative obeys $[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}$ and permits nonzero intertwined left/right curvatures. Lorentz-slice and metric compatibility reduce its pure no-new-field representative to $A_\mu = \Omega_\mu$, $B_\mu = -\Omega_\mu^\ddagger$, modulo a cancelling central term. We then prove that this pure pairing with $K = 0$ makes every nondegenerate generated tetrad admit a concurrent vector $\mathring{\nabla}_\mu V^\nu = {\delta_\mu}^\nu$; it therefore excludes the non-flat Schwarzschild vacuum exterior with nonzero mass and generic torsion-free Einstein geometries. This no-go is specifically torsion-free. Conversely, on every sufficiently small non-null Gaussian patch we construct an explicit metric-compatible composite contortion and a Lorentz-real single-Θ field satisfying $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$ for an arbitrary smooth tetrad. Thus local curved representability is closed without independent A_μ, B_μ fields. At the dynamical level we derive the exact fixed-background spin current of the paired kinetic term and prove two complementary statements: the effective Palatini variation of the minimal first-order branch excludes every flat affine representer, for every symmetric Lorentz-invariant slice pairing (which is unique up to scale).

The auxiliary exact-gradient restriction $e^a = \mathcal{N}_0^{-1/2} dY^a$ does admit the affine representer, but for a stronger and non-dynamical reason: every nondegenerate metric in that restriction is locally a pullback of Minkowski space, so its curvature and Hilbert–Palatini density vanish identically and the remaining action is a Jacobian null Lagrangian. The canonical self-consistent D-composite variation, action selection, physical torsion constraints, global continuation, and Einstein dynamics remain open.

Contents

1 Introduction: UBT and the role of Θ

Unified Biquaternion Theory is a mathematical-physics research programme that asks whether spacetime geometry and internal field structure can be encoded in one algebra-valued dynamical object rather than introduced as unrelated fundamental sectors. Its canonical starting point is the map

$$\boxed{\Theta : \mathcal{M} \times \mathbb{C}_\tau \longrightarrow \mathbb{B} = \mathbb{C} \otimes \mathbb{H}, \qquad \tau = t + i\psi,} \tag{1}$$

where $q \in \mathcal{M}$ denotes a spacetime point, t is physical time, and ψ is the additional imaginary component of the canonical complex-time argument. At each (q, τ), the value of the field is a biquaternion,

$$\Theta(q, \tau) = \theta^0(q, \tau)\mathbf{1} + \theta^k(q, \tau)\mathbf{e}_k, \qquad \theta^a \in \mathbb{C}, \tag{2}$$

so the minimal field value has four complex, or eight real, components. The commuting complex unit i and the quaternion units $\mathbf{e}_k$ play different algebraic roles. Their combination supplies complex phase freedom together with noncommutative quaternionic multiplication and supports the Lorentz-spin and left/right actions used below. The symbol Θ names the UBT master field; it does not by itself assert that the field is a Jacobi theta function. Such functional ansätze may be studied in reduced or spectral models, but no Jacobi-theta assumption is used in the geometric results of this paper.

Calling Θ "fundamental" means that the canonical programme does not postulate an independent spacetime metric as a second starting object. It also requires any connection, torsion, matter multiplet, or gauge structure used in a completed theory to be derived from Θ or justified as a composite or auxiliary nonpropagating variable. This is an architectural postulate, not a claim that all of those derivations have already been completed.

In the gravitational sector studied here, the value of Θ is not itself the metric. The local geometric carrier is its covariant first jet,

$$E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta. \tag{3}$$

When the four E_μ lie in the real Lorentz slice of $\mathbb{B}$ and are nondegenerate, they form a tetrad. Their complete symmetrized biquaternionic product is central and defines the metric,

$$\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = g_{\mu\nu}\mathbf{1}. \tag{4}$$

The conceptual chain investigated in this article is therefore

$$\boxed{\Theta \;\longrightarrow\; D_\mu\Theta \;\longrightarrow\; E_\mu \;\longrightarrow\; (g_{\mu\nu}, \omega, R).} \tag{5}$$

The first arrow depends on the representation and connection acting on Θ; the second identifies the tetrad; the last two are the standard metric, connection, and curvature constructions once the tetrad and torsion are specified.

The wider UBT programme also contains proposed gauge, quantum, spectral, and complex-time research tracks. They are outside the proof target of this paper. The present question is deliberately narrower: whether the single-field ansatz supplies a mathematically consistent local Lorentzian geometry, which parts of the GR tetrad chain follow kinematically, and which steps still require a canonical action and dynamical proof. In particular, this article does not claim a completed derivation of Einstein dynamics or of the Standard Model.

2 Scope and proof status of this paper

The purpose of this paper is to establish a minimal local geometric structure consistent with the original biquaternionic intuition of UBT and to identify exactly what remains to be proved. We do not claim an unconditional derivation of the Einstein equations.

Proved in this draft

- a projection-free Lorentzian metric from a central biquaternionic anticommutator;
- local representation of every Lorentz metric and rank ten of the tetrad-to-metric differential, with six Lorentz gauge directions;
- unique reconstruction of every metric-compatible frame connection from the tetrad and specified torsion;
- the torsion-free Levi–Civita spin connection as the unique classical GR connection;
- an explicit affine Θ representer of every constant Lorentz tetrad, including Minkowski spacetime;
- a no-go theorem for the naive one-sided invertible curved branch and the exact curvature identity for a two-sided biquaternionic derivative;
- exact reduction of the pure Lorentz-compatible pair to one spin connection, with no independent fields A_μ, B_μ;
- a concurrent-vector no-go theorem for the torsion-free pure pair, excluding the non-flat Schwarzschild vacuum exterior with nonzero mass;
- an explicit local torsionful single-Θ representer for every smooth Lorentzian tetrad, with composite metric-compatible contortion and no independent A_μ, B_μ fields;
- the exact fixed-background spin current of the paired kinetic term and a flat affine no-go for the effective Palatini variation of the minimal first-order branch;
- uniqueness (up to scale) of the symmetric Lorentz-invariant slice pairing, so pairing selection cannot remove that obstruction;
- an exact-gradient flatness no-go: if $e^a = \mathcal{N}_0^{-1/2} dY^a$, every nondegenerate induced metric is locally flat; affine stationarity for all coefficients is only the resulting Jacobian/null-Lagrangian corollary, not a curved-GR branch.

Open

- deriving vacuum torsion or spin-sourced torsion from the canonical UBT action;
- deriving from the canonical action whether the local composite-contortion branch is selected, and proving that its torsion is auxiliary/nonpropagating and physically admissible;
- alternatively, deriving a torsion-free relative left/right action and proving that it is composite or auxiliary rather than propagating;
- proving global existence and continuation of the curved-space system through zeros of V^2, Gaussian caustics, horizons, and topology;
- proving that the complete dynamics preserves the real Lorentz slice;
- deriving Einstein dynamics and the perturbation bridge from the original UBT master dynamics;
- canonical on-shell generation of Schwarzschild, Kerr, FRW, and wave sectors.

3 Relation to prior work

Standard machinery, not claimed as new

The tetrad postulate, Levi–Civita spin connection, torsion, and contorsion used below are standard Cartan/tetrad geometry [1, 2]. Treating gravity as a gauge theory of the Lorentz or Poincaré group goes back to Utiyama and Kibble [6, 7]; self-dual $\mathfrak{sl}(2, \mathbb{C})$ connection variables are central to the Ashtekar formulation [8]. None of this frame machinery is claimed to originate here.

Convergent programmes

Complex and Hermitian extensions of the metric were pursued by Einstein [9] and revived as nonsymmetric gravitation by Moffat [10]; complexified gravity on noncommutative spaces appears in Chamseddine [4]. Formulating physics, including gravity, inside a fixed geometric algebra on a flat background is the programme of spacetime algebra and gauge theory gravity [11, 12]. Teleparallel gravity demonstrates that the carrier of gravitation may be the anholonomy of a frame field rather than metric curvature [13]. Square roots of the metric as fundamental variables were analyzed by Ogievetskiĭ and Polubarinov [14]. Biquaternionic formulations of relativistic dynamics include de Haas [5], whose recent gravitational-rotor construction [15] generates the metric directly from the biquaternion algebra through the adjoint action of a $\mathrm{Spin}(1, 3)$ rotor and recovers Painlevé–Gullstrand forms of standard geometries. We regard the present framework as structurally convergent with these programmes, not derived from them, and make no priority claim over any of the mechanisms just listed.

Delimitation of what is claimed as new

The specific contributions claimed here, each stated as a theorem with a machine-verified proof, are: the projection-free central Jordan identity — the Lorentzian metric as the central adjugate anticommutator $\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1}$ on the Lorentz slice, with no trace, real-part map, or fiber average — together with its rank-ten theorem; the two-sided bimodule derivative with the exact pure-pair reduction $A_\mu = \Omega_\mu$, $B_\mu = -\Omega_\mu^\ddagger$; the concurrent-vector no-go for the torsion-free pure pair; the composite-contortion local representer for arbitrary smooth tetrads; and the dynamical-level results of the torsion-dynamics section (fixed-background spin current, affine and pairing no-go theorems, exact-gradient flatness, and the linearized D-composite symbol identity with its six-dimensional exponential sector). This delimitation is deliberate: the mechanism by which the metric arises differs from the rotor adjoint action of Ref. [15] (no group element, no Dirac basis; the metric is a bilinear central invariant of the covariant jet of a single field), and the negative results have, to our knowledge, no analogue in the cited programmes. A systematic prior-art comparison table is maintained in the repository.

4 Biquaternionic algebra and the Lorentz slice

Let $\mathbb{B} = \mathbb{C} \otimes \mathbb{H}$, with commuting complex unit i and quaternion units

$$\mathbf{e}_j \mathbf{e}_k = -\delta_{jk}\mathbf{1} + \epsilon_{jk\ell}\mathbf{e}_\ell.$$

Quaternion conjugation $\sharp$ reverses the quaternion units without conjugating the commuting complex coefficients:

$$(z^0\mathbf{1} + z^k\mathbf{e}_k)^\sharp = z^0\mathbf{1} - z^k\mathbf{e}_k.$$

Definition 4.1 (Classical Lorentz module)**.**

$$W_L = \{ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R}\}.$$

For $X = ix^0\mathbf{1} + x^k\mathbf{e}_k$ *and* $Y = iy^0\mathbf{1} + y^k\mathbf{e}_k$*, let*

$$\eta(X, Y) = -x^0y^0 + x^ky^k.$$

Theorem 4.2 (Central anticommutator identity)**.** *For all* $X, Y \in W_L$*,*

$$\boxed{\frac{1}{2}(X^\sharp Y + Y^\sharp X) = \eta(X, Y)\mathbf{1}.} \tag{6}$$

Proof. In scalar–vector notation, quaternion multiplication is $(a, \mathbf{u})(b, \mathbf{v}) = (ab - \mathbf{u}\cdot\mathbf{v}, a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v})$. For $X = (ix^0, \mathbf{x})$, $X^\sharp = (ix^0, -\mathbf{x})$. The scalar part of $X^\sharp Y$ is $-x^0y^0 + \mathbf{x} \cdot \mathbf{y}$. The vector parts of $X^\sharp Y$ and $Y^\sharp X$ are opposites, including the cross products, so they cancel in the half-sum. □

5 The covariant UBT tetrad

The fundamental field remains

$$\Theta(q, \tau) \in \mathbb{B}, \qquad \tau = t + i\psi.$$

On physical spacetime it is assumed smooth. Define

$$\boxed{E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta,} \tag{7}$$

where $\mathcal{N}_0 > 0$ is fixed globally and D_μ is a covariant derivative. The connection is kept representation-independent at this stage:

$$D_\mu \Theta = \partial_\mu \Theta + \rho_*(\Omega_\mu)\Theta. \tag{8}$$

The map ρ_* must be specified in each concrete representation; Eq. (8) does not assume left, right, or two-sided action prematurely.

Definition 5.1 (Admissible classical tetrad)**.** *A configuration belongs to the classical tetrad sector if*

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k \in W_L$$

for all μ and the matrix $e_\mu{}^a$ is invertible.

6 Projection-Free Metric and Algebraic Bivector

By Theorem 1,

$$\boxed{\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1},} \tag{9}$$

where

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1). \tag{10}$$

The complete left-hand side is already a central real element. Hence Eq. (9) defines the metric without discarding or projecting noncentral components. The fixed constant $\mathcal{N}_0$ only sets units.

The complementary antisymmetric product is

$$\Sigma_{\mu\nu} := \frac{1}{2}(E_\mu^\sharp E_\nu - E_\nu^\sharp E_\mu), \qquad \Sigma_{\nu\mu} = -\Sigma_{\mu\nu}. \tag{11}$$

Thus one product contains both the symmetric central geometry and an oriented bivector sector. No gauge-field identification of $\Sigma_{\mu\nu}$ is made without a separate transformation and dynamical analysis.

7 The local rank theorem

Define $\Phi(e) = e\eta e^\mathsf{T}$.

Theorem 7.1 (Full local metric rank)**.** *At every invertible tetrad, the differential $d\Phi_e : \mathbb{R}^{4\times 4} \to \mathrm{Sym}_4(\mathbb{R})$ is surjective and has rank ten. Its kernel is the six-dimensional local Lorentz algebra.*

Proof. The first variation is

$$\delta g_{\mu\nu} = \eta_{ab}(\delta e_\mu{}^a e_\nu{}^b + e_\mu{}^a \delta e_\nu{}^b).$$

For an arbitrary symmetric $h_{\mu\nu}$ choose

$$\delta e_\mu{}^a = \frac{1}{2} h_{\mu\rho} e^{\rho a}.$$

Then $\delta g_{\mu\nu} = h_{\mu\nu}$. Hence the differential is surjective and has rank ten. The domain has dimension sixteen, leaving a six-dimensional kernel. The transformations $\delta e = e\lambda$ with $\lambda^\mathsf{T}\eta + \eta\lambda = 0$ lie in the kernel and constitute the six-dimensional Lorentz Lie algebra. $\square$

Corollary 7.2. *The local comparison "a biquaternion has eight real components, whereas a metric has ten" is not the relevant rank count. The metric depends on four covariant first derivatives, which form a sixteen-component real tetrad before local Lorentz redundancy is removed.*

This theorem closes the local algebraic rank question. It does not prove that all tetrad variations arise from admissible variations of one on-shell Θ and its derived connection.

8 Connection and Christoffel symbols

The Christoffel symbols and the frame connection describe different indices:

- $\Gamma^\rho{}_{\mu\nu}$ transports coordinate indices;
- $\omega_\mu{}^a{}_b$, and its spin/biquaternionic lift Ω_μ, transport the local Lorentz frame.

They are related by

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu} e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0. \tag{12}$$

It is generally incorrect to identify Γ with a real component of Ω.

Theorem 8.1 (Unique torsion-free classical frame connection)**.** *For every nondegenerate C^2 tetrad, metric compatibility and vanishing torsion determine a unique Lorentz connection,*

$$\boxed{\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g) e_\rho{}^a - \partial_\mu e_\nu{}^a \right),} \tag{13}$$

with $\mathring{\omega}_{\mu ab} = -\mathring{\omega}_{\mu ba}$. Its spin lift is unique in the chosen representation, up to ordinary local Lorentz gauge.

Proof. The Levi–Civita connection $\mathring{\Gamma}(g)$ is uniquely fixed by $\mathring{\nabla} g = 0$ and $\mathring{\Gamma}^\rho{}_{[\mu\nu]} = 0$. Solving Eq. (12) gives Eq. (13). If two Lorentz connections have the same zero torsion, their difference is antisymmetric in its first two frame indices and symmetric in the indices forced by equality of torsion; cycling these symmetries gives zero. $\square$

The torsion-free result extends to arbitrary specified torsion. Write

$$T^a = de^a + \omega^a{}_b \wedge e^b = \frac{1}{2} T^a{}_{bc} e^b \wedge e^c. \tag{14}$$

Theorem 8.2 (Tetrad–torsion reconstruction)**.** *For every nondegenerate tetrad and specified torsion, there is exactly one metric-compatible Lorentz connection,*

$$\boxed{\omega^a{}_b = \mathring{\omega}^a{}_b(e) + K^a{}_b(T),} \tag{15}$$

where, with the convention above,

$$\boxed{K_{abc} = \frac{1}{2}\left(T_{cab} - T_{abc} - T_{bca}\right).} \tag{16}$$

Proof. Metric compatibility makes K_{abc} antisymmetric in a, b. Since the Levi–Civita part has zero torsion, $T_{abc} = K_{acb} - K_{abc}$. Combining this equation with its cyclic permutations yields Eq. (16), proving existence and uniqueness. $\square$

Thus Ω_μ is not an additional arbitrary classical matter field once the tetrad and torsion are fixed. The full-UBT problem is to derive T from the action. In the GR vacuum branch $T = 0$, so $K = 0$. Because every metric-compatible ω_μ lies in $\mathfrak{so}(1,3)$, its parallel transport preserves η_{ab} and the Lorentz slice, including in branches with nonzero contorsion.

The detailed proof and exact symbolic checks are in `canonical/gr_closure/gap_10omega_connection_elimination.tex` and `tools/verify_gap_10omega_connection.py`.

9 Curvature and special relativity

For a one-sided representation, the curvature is

$$\mathcal{R}_{\mu\nu} = \partial_\mu \Omega_\nu - \partial_\nu \Omega_\mu + [\Omega_\mu, \Omega_\nu]. \tag{17}$$

It must be distinguished from the algebraic tetrad bivector $\Sigma_{\mu\nu}$.

In a flat simply connected region, zero curvature permits a local frame with $\Omega_\mu = 0$. In inertial Cartesian coordinates one can also set $\Gamma^\rho{}_{\mu\nu} = 0$. The flat limit has an explicit single-field representer.

Theorem 9.1 (Affine representer of a constant Lorentz tetrad)**.** *Let $\bar{E}_\mu \in W_L$ be constant and linearly independent. Then*

$$\boxed{\Theta_{\text{aff}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\, \bar{E}_\mu x^\mu} \tag{18}$$

in the inertial gauge satisfies $\mathcal{N}_0^{-1/2} \partial_\mu \Theta_{\text{aff}} = \bar{E}_\mu$ and induces the constant metric determined by the central anticommutator.

For

$$\bar{E}_0 = i\mathbf{1}, \qquad \bar{E}_k = \mathbf{e}_k, \tag{19}$$

Eq. (18) becomes

$$\boxed{\Theta_{\text{SR}} = \Theta_0 + \sqrt{\mathcal{N}_0}\left(ix^0\mathbf{1} + x^k \mathbf{e}_k\right),} \tag{20}$$

which generates $g_{\mu\nu} = \eta_{\mu\nu}$. Every second spacetime derivative vanishes, so the standard flat source-free constant-coefficient second-order master operator annihilates Θ_{SR}. Nonzero connection coefficients in curvilinear coordinates do not by themselves imply curvature.

10 Integrability and representation selection

The tetrad is not independent in UBT; it must satisfy $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$. Once the classical connection is reconstructed from E and torsion, this becomes an implicit nonlinear first-order system in which the unknown tetrad occurs on both sides.

10.1 One-sided flatness obstruction

Suppose

$$D_\mu^L\Theta = \partial_\mu\Theta + A_\mu\Theta, \qquad [D_\mu^L, D_\nu^L]\Theta = F_{\mu\nu}^L\Theta. \tag{21}$$

Theorem 10.1 (One-sided invertible-field no-go)**.** *Assume $E_\nu = \mathcal{N}_0^{-1/2} D_\nu^L\Theta$, torsion-free antisymmetric tetrad compatibility, the same induced one-sided connection on $D_\nu\Theta$, and invertible Θ. Then $F_{\mu\nu}^L = 0$.*

Proof. Torsion-free compatibility makes the antisymmetrised covariant Hessian vanish, so $F_{\mu\nu}^L\Theta = 0$. Right multiplication by Θ^{-1} gives $F_{\mu\nu}^L = 0$. □

Thus the naive one-sided regular representation cannot be the generic curved GR route for invertible Θ. A noninvertible stabilizer sector is a possible exceptional branch, not a representation of arbitrary geometries.

10.2 Two-sided biquaternionic derivative

The algebra $\mathbb{B}$ is naturally a bimodule over itself. Consider

$$\boxed{D_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu.} \tag{22}$$

A direct expansion gives

$$\boxed{[D_\mu, D_\nu]\Theta = F_{\mu\nu}^A\Theta - \Theta F_{\mu\nu}^B.} \tag{23}$$

Torsion-free antisymmetric compatibility therefore reduces to

$$F_{\mu\nu}^A\Theta = \Theta F_{\mu\nu}^B. \tag{24}$$

For invertible Θ,

$$\boxed{F_{\mu\nu}^A = \Theta F_{\mu\nu}^B\Theta^{-1}.} \tag{25}$$

The left and right curvatures may be nonzero. The field Θ acts as an intertwiner between them. This does not prove existence for an arbitrary curved tetrad; it only exhibits an algebra-native escape from the stated one-sided flatness obstruction.

10.3 Pure Lorentz-compatible pair and no-new-field reduction

Let $\ddagger$ denote Hermitian conjugation and $W_L = \{X \in \mathbb{B} \mid X^\ddagger = -X\}$. Requiring Eq. (22) to preserve W_L for every $X \in W_L$ is equivalent to

$$B_\mu = -A_\mu^\ddagger + c_\mu\mathbf{1}, \qquad c_\mu \in \mathbb{R}. \tag{26}$$

If the same transport preserves the Lorentz quadratic form, the real dilation vanishes. Splitting off the trace then gives, modulo a common central one-form that cancels identically,

$$\boxed{A_\mu = \Omega_\mu, \qquad B_\mu = -\Omega_\mu^\ddagger,} \tag{27}$$

so that

$$D_\mu X = \partial_\mu X + \Omega_\mu X + X\Omega_\mu^\ddagger. \tag{28}$$

Thus A_μ and B_μ are not two new gravitational fields in the pure Lorentz branch. They are the left and right module actions of the one spin connection reconstructed from tetrad and torsion.

Theorem 10.2 (Concurrent-vector obstruction)**.** *Assume Eq. (28), zero torsion, and a nondegenerate tetrad satisfying $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$. Then the Lorentz-real projection of Θ defines a spacetime vector V such that*

$$\boxed{\nabla_\mu V^\nu = \delta_\mu{}^\nu.} \tag{29}$$

Consequently $\mathcal{L}_V g = 2g$ and $R^\rho{}_{\sigma\mu\nu}V^\sigma = 0$.

Proof. The derivative in Eq. (28) commutes with the Lorentz involution $\mathcal{J}X = -X^\ddagger$. Hence $X = (\Theta + \mathcal{J}\Theta)/2 \in W_L$ obeys $D_\mu X = \sqrt{\mathcal{N}_0}E_\mu$. Writing $X = X^a\mathbf{u}_a$, $E_\mu = e_\mu{}^a\mathbf{u}_a$ and defining $V^\nu = \mathcal{N}_0^{-1/2}e_a{}^\nu X^a$, the inverse tetrad postulate gives

$$\nabla_\mu V^\nu = \mathcal{N}_0^{-1/2}e_a{}^\nu D_\mu X^a = e_a{}^\nu e_\mu{}^a = \delta_\mu{}^\nu.$$

The homothety and curvature-kernel identities follow by symmetrising and antisymmetrising covariant derivatives. □

For the non-flat Schwarzschild vacuum exterior $ds^2 = -fdt^2 + f^{-1}dr^2 + r^2d\Omega^2$ with $f = 1 - 2M/r$, rotational averaging reduces any putative homothetic vector to $V = T(t,r)\partial_t + R(t,r)\partial_r$. The angular equation gives $R = r$, while the rr equation gives $f'(r) = 0$, contradicting $f' = 2M/r^2$ for $M \neq 0$. Therefore the pure pair is closed as a kinematic no-new-field reduction and its $K = 0$ branch is closed as a torsion-free no-go. This statement concerns the vacuum exterior geometry; it does not depend on whether an exactly nonrotating astrophysical black hole is realised.

10.4 Composite-contortion local curved representer

The torsion-free qualifier is essential. Let ρ be a shifted non-null Gaussian coordinate on a sufficiently small patch,

$$g^{\mu\nu}\partial_\mu\rho\partial_\nu\rho = \varepsilon, \qquad \varepsilon = \pm 1, \qquad \rho \neq 0, \tag{30}$$

and define

$$V^\mu = \varepsilon\rho\nabla^\mu\rho. \tag{31}$$

Then $V^2 = \varepsilon\rho^2$ and $V_\mu = \frac{1}{2}\partial_\mu(V^2)$. Put

$$W_{\mu\nu} = g_{\mu\nu} - \mathring{\nabla}_\mu V_\nu, \qquad \boxed{K_{\nu\mu\rho} = \frac{W_{\mu\nu}V_\rho - V_\nu W_{\mu\rho}}{V^2}.} \tag{32}$$

Because $W_{\mu\nu}V^\nu = 0$, the tensor in Eq. (32) obeys $K_{\nu\mu\rho} = -K_{\rho\mu\nu}$ and is therefore a metric-compatible contortion. Moreover,

$$K^\nu{}_{\mu\rho}V^\rho = W_\mu{}^\nu, \qquad \nabla_\mu^{(\mathring{\Gamma}+K)}V^\nu = \delta_\mu{}^\nu. \tag{33}$$

Let $V^a = e^a{}_\nu V^\nu$ and define

$$\boxed{\Theta = \sqrt{\mathcal{N}_0}\, V^a\mathbf{u}_a.} \tag{34}$$

The tetrad postulate for $\omega(e,K)$ then gives

$$D_\mu\Theta = \sqrt{\mathcal{N}_0}\, e^a{}_\nu\nabla_\mu^{(\mathring{\Gamma}+K)}V^\nu\mathbf{u}_a = \sqrt{\mathcal{N}_0}E_\mu. \tag{35}$$

Thus every smooth Lorentzian tetrad has a local single-Θ representer inside the same one-connection pairing $A=\Omega(e,K)$, $B=-\Omega(e,K)^{\ddagger}$. No independent A_μ or B_μ field is introduced. On a Schwarzschild exterior patch $r>2M$, one may take

$$\rho(r)=\rho_0+\int_{r_0}^{r}\frac{du}{\sqrt{1-2M/u}},\qquad V=\rho\sqrt{1-2M/r}\,\partial_r. \tag{36}$$

This explicitly represents the non-flat vacuum exterior locally, but with generally nonzero composite torsion.

A relative pair may still be parameterised as

$$A_\mu=\Omega_\mu+P_\mu,\qquad B_\mu=-\Omega_\mu^{\ddagger}+Q_\mu. \tag{37}$$

It is no longer required for local kinematic curved representability, but it remains a possible torsion-free completion. If used, the relative action $P_\mu X-XQ_\mu$ must be derived from the canonical action and shown composite or auxiliary rather than propagating. A common central constant or one-form cannot remove the torsion-free obstruction because it cancels from the derivative.

The resulting equation is schematically

$$E_\mu=\mathcal{N}_0^{-1/2}\left[\partial_\mu\Theta+A_\mu[E,T]\Theta-\Theta B_\mu[E,T]\right]. \tag{38}$$

It is an implicit nonlinear PDE/fixed-point system. If Θ is restricted to a Jacobi-theta function class, the complete system may additionally be transcendental; implicitness and transcendental functional dependence are separate mathematical properties.

The one-sided/two-sided integrability proof is in `canonical/gr_closure/gap_10i_integrability_selection.tex`. The pure-pair reduction and torsion-free concurrent-vector/Schwarzschild-exterior no-go are in `canonical/gr_closure/gap_10i_paired_connection_audit.tex` and `tools/verify_gap_10i_paired_connection.py`. The local composite-contortion right inverse and its exact verifier are in `canonical/gr_closure/gap_10i_torsionful_local_representer.tex` and `tools/verify_gap_10i_torsionful_local_representer.py`.

10.5 Exact prescribed-connection integrability criterion

For specified smooth (E_μ,A_μ,B_μ), the inhomogeneous system can be converted into a homogeneous parallel-section problem. Let $\mathcal{C}_\mu X=A_\mu X-XB_\mu$, $s=\sqrt{\mathcal{N}_0}$, and

$$Y=\begin{pmatrix}\Theta\\1\end{pmatrix},\qquad \widehat{\mathcal{C}}_\mu=\begin{pmatrix}\mathcal{C}_\mu & -sE_\mu\\0&0\end{pmatrix}. \tag{39}$$

Then $D_\mu\Theta=sE_\mu$ is exactly $(\partial_\mu+\widehat{\mathcal{C}}_\mu)Y=0$. Its curvature is

$$\widehat{\mathcal{F}}_{\mu\nu}=\begin{pmatrix}\mathcal{F}_{\mu\nu} & -s(\mathcal{D}_\mu E_\nu-\mathcal{D}_\nu E_\mu)\\0&0\end{pmatrix}, \tag{40}$$

where $\mathcal{F}_{\mu\nu}X=F^A_{\mu\nu}X-XF^B_{\mu\nu}$. For an initial value $Y_0=(\Theta_0,1)^T$, a single-valued solution exists on a connected domain exactly when the augmented holonomy fixes Y_0; on a simply connected domain, flat augmented curvature is a sufficient condition for every initial value. Thus the prescribed-coefficient integrability problem is closed. GAP-10I-CURVED now has local kinematic existence closed by the composite-contortion construction. What remains is the UBT-specific canonical selection of (E,A,B,T), physical torsion constraints, regularity and global continuation. The detailed proof is in `canonical/gr_closure/gap_10i_augmented_holonomy.tex`.

11 Torsion dynamics in the minimal first-order branch

Consider the standard Hilbert–Palatini action with independent tetrad and Lorentz connection,

$$S_{\mathrm{HP}} = \frac{1}{4\kappa} \int \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}(\omega) + S_{\mathrm{m}}[e, \omega, \Theta], \tag{41}$$

and define $\delta_\omega S_{\mathrm{m}} = \frac{1}{2} \int \tau_{ab} \wedge \delta\omega^{ab}$. Connection variation gives the algebraic Cartan equation

$$\boxed{\epsilon_{abcd} T^a \wedge e^b = \kappa \tau_{cd}.} \tag{42}$$

For a nondegenerate tetrad the 24-component map $T^a \mapsto \epsilon_{abcd} T^a \wedge e^b$ has rank 24. Hence zero spin current forces $T = 0$, while a specified spin current determines a unique contorsion. This closes GAP-10T-PALATINI conditionally. The direct tree-level matter spin current of the paired kinetic term has now been derived in the effective Palatini variation with e, g, the volume form and Θ held fixed. For an affine flat representer it is nonzero away from at most one point, so the minimal Hilbert–Palatini plus kinetic branch does not admit the flat inertial torsion-free solution. Moreover, the $\sharp$/Minkowski pairing is unique up to scale among real symmetric Lorentz-invariant slice pairings; the $\ddagger$ Hilbert–Schmidt pairing fails boost invariance. Hence pairing selection alone cannot remove the obstruction. The auxiliary exact-gradient restriction, with $e^a = \mathcal{N}_0^{-1/2} dY^a$, does admit the affine representer for all values of Λ, κ, and $\mathcal{N}_0$. However, this is not a surviving curved branch: nondegeneracy makes Y^a local coordinates and $g = \mathcal{N}_0^{-1} Y^* \eta$, so the Riemann curvature and Hilbert–Palatini density vanish identically for every configuration of the restriction. The locked kinetic and cosmological terms reduce to a Jacobian null Lagrangian. The canonical self-consistent D-composite variation, the origin and normalization of the Hilbert–Palatini term, and curved dynamics remain open; Finally, the last remaining scheme — the self-consistent D-composite variation, with the connection inside the defining jet — has been analyzed at the linearized, frozen-coefficient level in the Lorentz-real W_L subsector. The frozen symbol satisfies the exact operator identity $A^3 = q\,A^2$ with $q = \lambda \cdot s$, giving $\det(I - A) = (1 - q)^6$; off $q = 1$ every $\delta\Theta$-driven mode is an exact gradient (pullback-flat), while at $q = 1$ there is an exactly six-dimensional sector of anholonomic modes carrying nonzero linearized Riemann image. For real Fourier frequencies q is imaginary, so this sector is of exponential/evanescent type and its relation to real-frequency propagation, its gauge and ghost content, and its quadratic action are open. Imposing $\Theta \in W_L$ is a consistent subsector, not proved necessary. For all these reasons GAP-10T-DYN is narrowed rather than fully closed. The proofs and exact verifiers are in `canonical/gr_closure/gap_10t_palatini_torsion_dynamics.tex`, `canonical/gr_closure/gap_10tdyn_10d_canonical_action_audit.tex`, `canonical/gr_closure/gap_10t_composite_flat_admissibility.tex`, `tools/verify_remaining_gr_subclosures.py`, and `tools/verify_canonical_spin_current.py`.

12 Lorentz-slice and imaginary-time stability

The real Lorentz slice has an intrinsic fixed-set characterization. Define

$$\mathcal{J}X = -\overline{X^\sharp}. \tag{43}$$

Then $\mathcal{J}$ is an anti-linear involution and $\mathrm{Fix}(\mathcal{J}) = W_L$. Consequently, every unique Cauchy evolution equivariant under $\mathcal{J}$, with $\mathcal{J}$-real sources and initial data, preserves the Lorentz slice by uniqueness. This closes GAP-10L-SYM conditionally; the canonical equations must still be shown to satisfy the equivariance and well-posedness hypotheses.

For imaginary time, a tetrad flow

$$\partial_\psi e_\mu{}^a = \Lambda_\psi{}^a{}_b e_\mu{}^b, \qquad \Lambda_{\psi ab} = -\Lambda_{\psi ba}, \tag{44}$$

is pure local Lorentz gauge and obeys $\partial_\psi g_{\mu\nu} = 0$ exactly. Strictly ψ-independent data are also preserved by every unique ψ-translation-invariant evolution. These results close GAP-10ψ-KIN and GAP-10ψ-SYM conditionally, while the overall physical stability problem remains narrowed. Details are in `canonical/gr_closure/gap_10l_psi_symmetry_propagation.tex`.

13 Conditional infrared endpoint of the metric dynamics

Variation of the minimal tetrad–Palatini action gives the Cartan equation and, in the zero-spin-current branch, Einstein's equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}. \tag{45}$$

Moreover, Lovelock uniqueness implies that a four-dimensional local natural symmetric divergence-free second-order metric equation with no additional light geometric fields must have the form $aG_{\mu\nu} + bg_{\mu\nu} = \kappa T_{\mu\nu}$. Thus the conditional low-energy endpoint is no longer ambiguous: under those assumptions it is Einstein–Λ. GAP-10D-PALATINI and GAP-10D-UNIQUENESS are therefore closed conditionally. GAP-10D itself remains narrowed because canonical UBT must still derive the low-energy assumptions, coefficients and matter action. The precise scope is stated in `canonical/gr_closure/gap_10d_low_energy_uniqueness.tex`.

14 Complex time and regularity

The field may depend on $\tau = t + i\psi$. The metric definition Eq. (9) is pointwise; it does not integrate over ψ. A classical four-dimensional sector may require

$$\partial_\psi g_{\mu\nu} = 0,$$

but this must follow from the dynamics or a clearly stated classical-sector condition.

Smoothness of the physical spacetime field and intrinsic biquaternionic regularity in the argument q are different properties. The latter has not yet been uniquely fixed: Cauchy–Fueter, slice-regular, and UBT-specific Jacobi structures must be compared before a regularity axiom is locked.

15 Relation to earlier compact-fiber closure

Earlier work defined a metric by averaging a quadratic form over the compact imaginary-time coordinate and used the resulting function space to repair a rank count. Those constructions remain mathematically auditable, but they are not required by the central anticommutator theorem and are no longer the canonical local metric definition. They are retained in the repository as an exploratory candidate-completion branch and are documented in `canonical/gr_closure/HISTORICAL_FIBER_ROUTE_STATUS.md`. The branch is rejected as canonical because of weak selection and large representer redundancy, not because its mathematics was disproved. In particular, the earlier fiber-average constructions are not used as a canonical derivation of the metric, rank closure, connection, or Einstein dynamics.

The conceptual distinction is:

- the old route enlarged the effective target by retaining a whole ψ profile;
- the present route uses the ordinary tetrad content of the covariant first jet and obtains $16 - 6 = 10$ locally.

16 Schwarzschild and perturbations: honest current status

The repository contains an explicit and numerically verified construction of the spatial isotropic Schwarzschild metric. The static vacuum Einstein relations determine the lapse uniquely once the spatial branch and boundary conditions are assumed. What remains open is the selection of the complete Schwarzschild tetrad by the canonical UBT master dynamics and connection.

The relation $\partial_\psi \Theta = i\Phi(r)\Theta$ is not used as a canonical derivation of the lapse, and a generic Maxwell or $U(1)_\psi$ field does not generate vacuum Schwarzschild. The perturbation bridge must be rederived in the covariant connection/tetrad formulation; earlier fiber-linearisation results remain conditional results of the exploratory branch.

17 Revised gap ledger

Gap	Status
GAP-10K	**CLOSED locally**: tetrad-to-metric differential has rank ten.
GAP-10Ω-KIN	**CLOSED**: specified tetrad and torsion uniquely determine the metric-compatible frame connection.
GAP-10Ω-GR	**CLOSED**: the torsion-free branch is the unique Levi–Civita spin connection, up to Lorentz gauge.
GAP-10T-PALATINI	**CLOSED CONDITIONALLY**: minimal first-order variation gives an invertible algebraic Cartan torsion equation.
GAP-10T-SPIN / FLAT-NOGO	**CLOSED CONDITIONALLY / CLOSED AS NO-GO**: direct fixed-background current derived; minimal affine torsion-free branch excluded.

Gap	Status
GAP-10T-PAIRING-NOGO	**CLOSED AS NO-GO**: the $\sharp$/Minkowski form is the unique nonzero symmetric Lorentz-invariant slice pairing up to scale, so pairing selection cannot cure the affine obstruction.
GAP-10T-GRADIENT-FLATNESS	**CLOSED AS NO-GO**: every nondegenerate exact-gradient tetrad $e^a = \mathcal{N}_0^{-1/2} dY^a$ induces a locally flat metric; affine stationarity is only a Jacobian/null-Lagrangian corollary.
GAP-10T-DCOMP-WL-SECTOR	**CLOSED CONDITIONALLY**: $\Theta \in W_L$ is a consistent Lorentz-real subsector; necessity disproved by a constant Hermitian counterexample.
GAP-10T-DCOMP-LIN-OFFRES	**CLOSED CONDITIONALLY**: frozen symbol obeys $A^3 = qA^2$; off $q = 1$ driven modes are exact gradients (pullback-flat), in the W_L exponential-symbol scheme.
GAP-10T-DCOMP-RES	**OPEN**: six-dimensional exponential symbol sector with nonzero linearized Riemann image; real-frequency relation, gauge/ghost content, and quadratic action undecided.
GAP-10T-DYN	**NARROWED**: decide the exponential sector (quadratic action, propagation), variable-coefficient assembly, on-shell torsion, and coefficient origin.
GAP-10L-CONN	**CLOSED**: metric-compatible Lorentz transport preserves η and the Lorentz slice.
GAP-10L-SYM	**CLOSED CONDITIONALLY**: unique $\mathcal{J}$-equivariant evolution preserves the Lorentz fixed set.
GAP-10L-DYN	**NARROWED**: verify equivariance and well-posed uniqueness for the complete UBT dynamics.
GAP-10I-SR	**CLOSED**: explicit affine Θ representer for every constant Lorentz tetrad.
GAP-10I-1S	**CLOSED AS NO-GO**: naive one-sided invertible torsion-free branch forces zero curvature.
GAP-10I-PAIR-KIN	**CLOSED**: Lorentz and metric compatibility reduce the pure pair to one spin connection modulo a cancelling central term.
GAP-10I-PAIR-GR	**CLOSED AS TORSION-FREE NO-GO**: with $K = 0$ the pure pair implies a concurrent vector and excludes the non-flat Schwarzschild vacuum exterior with $M \neq 0$.
GAP-10I-TORSION-LOCAL	**CLOSED LOCALLY**: explicit single-Θ representer for every smooth tetrad using composite metric-compatible contortion.

Gap	Status
GAP-10I-2S	**NOT REQUIRED FOR LOCAL KINEMATICS**: optional torsion-free composite/auxiliary route.
GAP-10I-PRESCRIBED	**CLOSED**: exact augmented-holonomy criterion for specified (E, A, B).
GAP-10I-CURVED	**LOCAL KINEMATICS CLOSED; DYNAMICS/GLOBAL NARROWED**: canonical selection, physical torsion constraints, regularity, and global continuation remain.
GAP-10D-PALATINI / UNIQUENESS	**CLOSED CONDITIONALLY**: minimal first-order dynamics and Lovelock infrared uniqueness.
GAP-10D	**NARROWED**: derive those low-energy assumptions, couplings, and matter sector from canonical UBT.
GAP-10ψ-KIN / SYM	**CLOSED / CLOSED CONDITIONALLY**: Lorentz-gauge flow or translation symmetry protects the metric.
GAP-10ψ	**NARROWED**: show which mechanism is selected and exclude unstable physical ψ modes.
GAP-B-MASTER	**OPEN**: perturbation bridge from the original master dynamics.
GAP-U2Θ	**OPEN**: canonical dynamical generation of the full Schwarzschild tetrad/lapse.

18 Conclusion

The clean local UBT geometry is

$$\boxed{\Theta \;\longrightarrow\; E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0} \;\longrightarrow\; \begin{cases} \frac{1}{2}\{E_\mu, E_\nu\}_\sharp = g_{\mu\nu}\mathbf{1}, \\ \frac{1}{2}[E_\mu, E_\nu]_\sharp = \Sigma_{\mu\nu}, \\ [D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu} \quad \text{in the general two-sided branch.} \end{cases}}$$

This structure resolves the local rank objection without auxiliary maps, metric projections, or fiber averaging. The connection is also no longer an arbitrary addition: for specified tetrad and torsion it is uniquely reconstructed, and the torsion-free GR branch is Levi–Civita. Special relativity has an explicit affine single-Θ representer. The naive one-sided curved route is ruled out for generic invertible Θ. The two-sided derivative supplies a nonflat intertwiner condition, while the pure Lorentz-compatible pair is reduced to the existing spin connection. Its $K = 0$ branch is ruled out for the non-flat Schwarzschild vacuum exterior by the concurrent-vector theorem, but an explicit composite contortion provides a local single-Θ right inverse for every smooth tetrad. The decisive remaining work is therefore dynamical rather than local kinematic: the fixed-background spin current is now derived and the effective Palatini scheme of the minimal branch is excluded at the flat point for every admissible pairing. The auxiliary exact-gradient restriction admits the flat representer only because it is locally flat for every nondegenerate configuration. The decisive tasks are to compute the canonical self-consistent D-composite variation and curved dynamics, establish algebraic nonpropagation and physical torsion bounds, derive the low-energy coefficients, prove global continuation, establish well-posed symmetry propagation, and generate observed geometries on shell.

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